

MMA League Table

Computer Science NEA

Thomas Jubb

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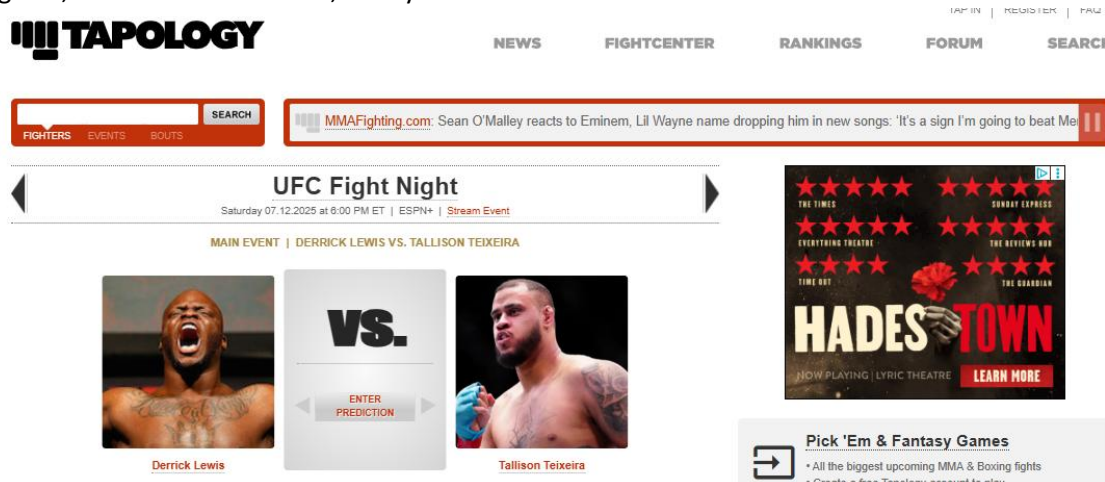
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Desk-based research

Example 1 – Tapology

Tapology is an MMA website which shows rankings, news and upcoming fights.

- The page shows which fights are about to happen, and how it also lets you enter a prediction.
- When the page shows what fights are about to happen, it also has two pictures of each of the fighters that will be fighting in the bout and their names.
- You can also search for individual fighters, bouts and events.
- The Tapology system shows the main pages of the website on top.
- In the rankings page, at the top, you are given the option to sort the rankings by 4 broad categories.
- The rankings page shows the major trending changes to the rankings at the top and shows a graph of the rankings change for all the fighters
- Below that, it shows the ranking changes to all the elite fighters.
- The website also shows the top 50 ranking changes in a table form
- The website shows individual ranking boards in a large table format
- The table has the ranking first, then a picture of the fighter. It then shows information about the fighter, and then their record, finally.



WORLDWIDE	REGIONAL	ALL TIME	ANNUAL	CUSTOM	MY RANKINGS
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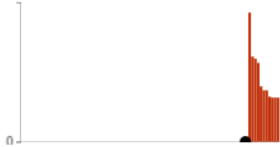
MAJOR TRENDING MEN'S CHANGES



38



 **Felipe Lima**
#40 Bantamweight
On June 28th Felipe lost to Payton Talbott by decision, unanimous at UFC 317 and fell to 14-2.




20



 **Zhalgas Zhumagulov**
#49 Flyweight
On June 28th Zhalgas defeated Beno Adamia by decision, unanimous at OKTAGON 73 and climbed to 18-9.

ELITE MEN'S CHANGES

  3  Paddy Pimblett #4 Lightweight	  2  Justin Gaethje #5 Lightweight
  1  Jailton Almeida #5 Heavyweight	  1  Merab Dvalishvili #1 Bantamweight
  1  Umar Nurmagomedov #2 Bantamweight	  1  Charles Oliveira #3 Lightweight

TOP 50 MEN'S CHANGES

	9	 Lucas Rocha	#53 Flyweight	down 9 from last week
	8	 CJ Vergara	#46 Flyweight	up 8 from last week
	7	 Jamall Emmers	#44 Featherweight	up 7 from last week
	7	 Eryk Anders	#40 Middleweight	up 7 from last week
	6	 Jarred Brooks	#59 Flyweight	down 6 from last week
	6	 Jon Jones	#29 Heavyweight	down 6 from last week
	6	 Derek Brunson	#43 Middleweight	down 6 from last week
	6	 Usman Nurmagomedov	#24 Lightweight	up 6 from last week
	6	 Rafa García	#39 Lightweight	up 6 from last week
	6	 Matt Frevola	#33 Lightweight	down 6 from last week

RANKINGS

1 2 3 4 5 ... Next › Last »

1



++

Ilia "El Matador" Topuria

17-0-0

Recent Fights:

TKO

TKO

TKO

DEC

SUB

TKO

TKO

2025

2024

2024

2023

2022

2022

2021

2



Islam Makhachev

27-1-0

Recent Fights:

SUB

SUB

TKO

DEC

SUB

TKO

SUB

2025

2024

2023

2023

2022

2022

2021

3



Alexander "The Great" Volkanovski

27-4-0

Recent Fights:

DEC

TKO

TKO

TKO

DEC

DEC

TKO

2025

2024

2023

2023

2023

2022

2022

4



Magomed Ankalaev

21-1-1, 1 NC

Recent Fights:

DEC

DEC

TKO

NC

DEC

TKO

DEC

2025

2024

2024

2023

2022

2022

2022

5



++

Merab "The Machine" Dvalishvili

20-4-0

Recent Fights:

SUB

DEC

DEC

DEC

DEC

DEC

TKO

2025

2025

2024

2024

2023

2022

2021

6



Alexandre "The Cannibal" Pantoja

30-5-0

Recent Fights:

SUB

SUB

DEC

DEC

DEC

SUB

SUB

2025

2024

2024

2023

2023

2022

2021

7



Jack Della Maddalena

18-2-0

Recent Fights:

DEC

TKO

DEC

DEC

SUB

TKO

TKO

2025

2024

2023

2023

2023

2022

2022

8



Dricus "Stillknocks" Du Plessis

23-2-0

Recent Fights:

DEC

SUB

DEC

TKO

TKO

SUB

DEC

2025

2024

2024

2023

2023

2022

2022

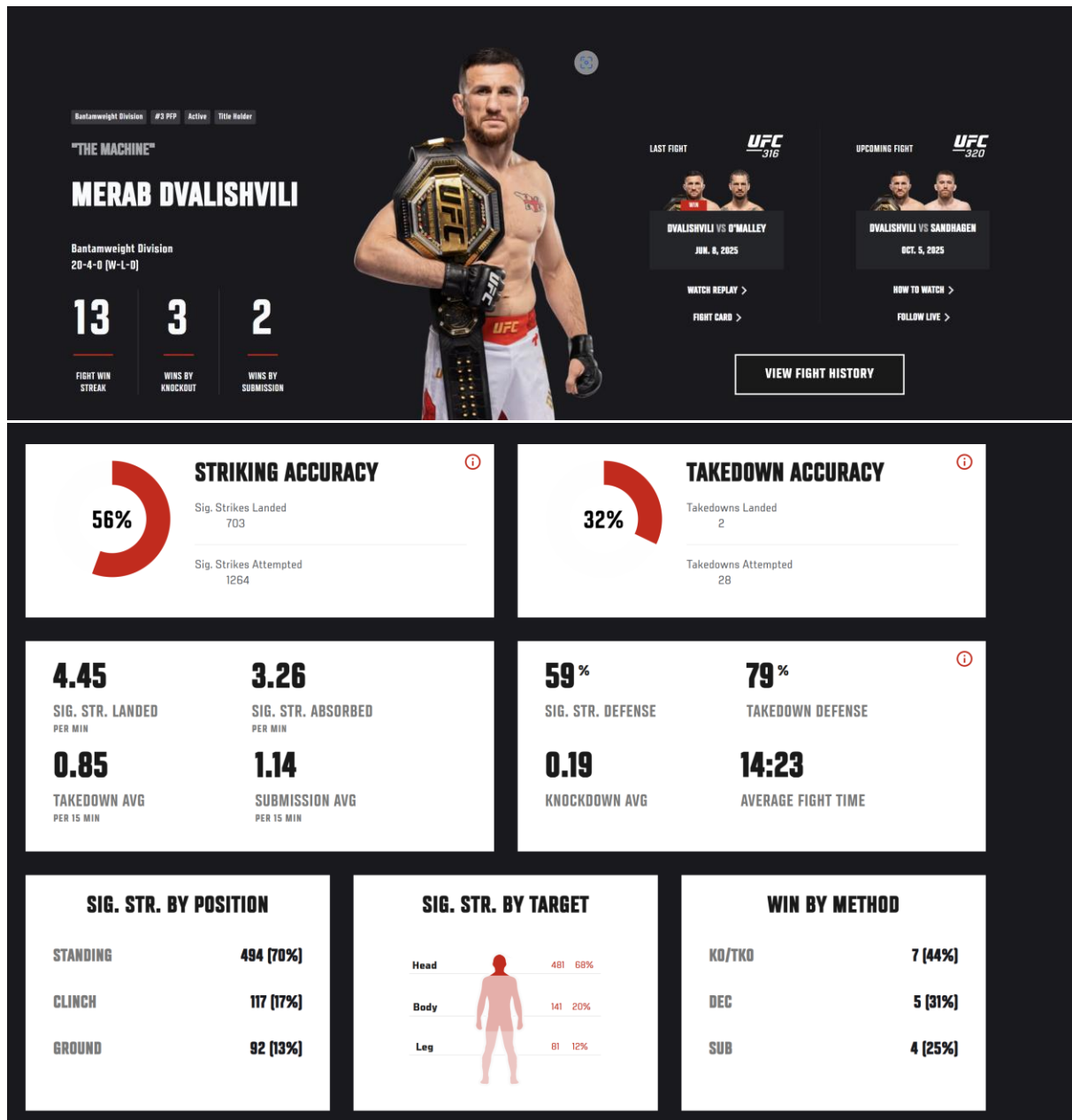
vs Khamzat Chimaev

Scheduled for Aug 18

- I think that it's a good idea to include the ability to search for individual fighters if you want to look at their stats. I don't really think that adding the bouts is necessary, as you'll see that information on the fighter's page anyway. Also, the system I'm making is focused on the fighters themselves, so I don't really think that I need the information about an event, either, meaning that I will add these
- I personally like this way of showing rankings in a large table format a lot. I think the large table format makes all the information stored way easier to digest.
- I may be limited in the way I can display lots of images at the same time from the SQL database
- I like the little flag which tells you which fighter is from which country.
- I don't like the order of the table. In my opinion, the name should be first, so you can read the names of each number ranking quicker, instead of having the picture of the fighter

- The front page of the UFC website shows every ranking with the champion's image at the top of each weight class ranking
- If you aren't the champion, the chart only shows your rank, name and rank change.
- For each fighter, it shows a picture of them, their division, their record, their win streak, the ways they've won their previous fights and their upcoming and last fights.
- It also shows tags which tell you other info, like if the fighter is a champion, if they're active
- Under that part, it shows 7 different boxes, which show different stats
- These stats come with charts a couple of times. For this example, there are two pie charts and heatmaps based on significant strikes

MEN'S POUND-FOR-POUND TOP RANK		FLYWEIGHT		BANTAMWEIGHT		FEATHERWEIGHT	
ILIA TOPURIA		ALEXANDRE PANTOJA		MERAB DVALISHVILI		ALEXANDER VOLKANOVSKI	
CHAMPION		CHAMPION		CHAMPION		CHAMPION	
							
1	Ilia Topuria	1	Joshua Van	1	Sean O'Malley	1	Movsar Evloev
2	Islam Makhachev	2	Brandon Moreno	2	Umar Nurmagomedov	2	Diego Lopes
3	Merab Dvalishvili	3	Brandon Royval	3	Petr Yan	3	Yair Rodriguez
4	Dricus Du Plessis	4	Amir Albazi	4	Cory Sandhagen	4	Brian Ortega
4	Alexandre Pantoja	5	Kai Kara-France	5	Song Yadong	5	Arnold Allen
6	Alexander Volkanovski	6	Tatsuro Taira	6	Deiveson Figueiredo	6	Lerone Murphy
7	Magomed Ankalaev	7	Manel Kape	7	Marlon Vera	7	Aljamain Sterling
8	Jack Della Maddalena	8	Alex Perez	8	Mario Bautista	8	Josh Emmett
9	Tom Aspinall	9	Asu Albabayev	9	Rob Font	9	Youssef Zalal
10	Alex Pereira	10	Steve Erceg	10	Aiemann Zahabi	10	Jean Silva
11	Max Holloway	11	Tim Elliott	11	Henry Cejudo	11	Patricio Pitbull
12	Belal Muhammad	12	Tagir Ulanbekov	12	Vinicius Oliveira	12	Steve Garcia
13	Arman Tsarukyan	13	Ramazan Temirov	13	Marcus McGhee	13	David Onama
14	Khamzat Chimaev	14	Bruno Silva	14	Kyler Phillips	14	Dan Ige



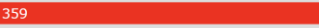
My takeaways

- I like how the front page shows all the rankings so compactly, but, in my opinion, they don't show enough information. I will have to find a balance between something like this and a UI that's too cluttered.
- I like the tag system; I can integrate this into SQL on fighter profiles.
- I like how the number of tables is more diverse than just having pie charts. In my project, I want to have a few more ways of showing the data to make the end user experience more interesting

Example 3 – FightMatrix

- The table is very simplistic and efficient

- It is limited in what it shows compared to the others
- The points it shows are fighter scores calculated using predictive data and past data

Rank	Fighter (Division)	Record	Points
1	 Dricus Du Plessis (MW)	23-2-0	370 
2	 Merab Dvalishvili (BW)	20-4-0	359 
3	 Islam Makhachev (LW)	27-1-0	346 
4	 Ilia Topuria (LW)	17-0-0	346 
5	 Alexandre Pantoja (FLW)	30-5-0	339 
6	 Magomed Ankalaev (LHW)	21-1-1	256 
7	 Brandon Moreno (FLW)	23-8-2	191 
8	 Jack Della Maddalena (WW)	18-2-0	132 
9	 Alexander Volkanovski (FW)	27-4-0	117 
10	 Alex Pereira (LHW)	12-3-0	107 
11	 Jiri Prochazka (LHW)	31-5-1	107 
12	 Joshua Van (FLW)	15-2-0	101 
13	 Movsar Evloev (FW)	19-0-0	97 
14	 Ciryl Gane (HW)	13-2-0	89 
15	 Tom Aspinall (HW)	15-3-0	89 

My Takeaways

- I like how simplistic the table is, as it lets you take in the essential information quickly
- I don't like how it doesn't show who the champion is because that is very important contextually to someone's ranking
- I don't think that it's necessary to show the point system directly to the user. I think that the ranking tells the user enough, and it takes up space where more important information can be.

Stakeholders

Fans

MMA Fans are the end-users who will look at the rankings and stats. They will also send requests to generate odds for fights. The fans are going to want a high-quality interface that is easy to use and visually pleasing. They are going to want quick access to specific fighter statistics to see a good enough picture in a short amount of time when they research a fighter. They want up-to-date and accurate statistics for fighters. Also, they want odds for fights that are as accurate as possible. They will also want extra features like voting predictions for fights, for example. I could collect this info by doing surveys of MMA fans to see what they would like out of an MMA system.

Fighters

MMA Fighters won't probably use the system directly, but they will want to have an accurate representation of the system out of fairness, meaning that they will want accurate rankings and accurate odds. The way I can meet the needs of fighters and appease them is by doing good research into odds, so I can find out the most accurate type, and making sure that the system I use to rank the fighters takes a lot of data into account and takes the data into account proportionately.

System Admins

The admins of a system are the ones who put the data into the system and host the server that the end-users access to see the fighter data. Like the end-users, they will want an easy-to-use and aesthetically pleasing interface for their part of the system, but they will also want the system to be efficient and reliable storage-wise. They will also want good features that can attract more users. I could get this data by interviewing an admin of an already existing MMA system to see what they like about the system.

Promoters

MMA Promoters are the people who usually run the championship or league that an MMA fighter will fight in. They will be the people who get the System Admins to run the system to show fighter data. They will want the end-user (MMA Fans) to have a good experience. This means up-to-date and accurate data and an easy-to-use interface. A promoter would also value things that make a fighter/bout look better or more entertaining, as this makes the event of an MMA fan paying for PPV or a ticket for an event a lot more likely to happen.

Investigation Method 1

I am researching MMA fans as they are the main users of my system. My goal for this part is to figure out what stats and features I can put into the frontend of the system to make it as user-friendly as possible. I chose to gather this data by using a questionnaire, as this is the most efficient way to get data from many people.

MMA Survey

Published

Questions
Responses
Settings

MMA Survey

B I U

MMA SURVEY

Rank These Qualities from 1 (least important) to 5 (most important)

Description (optional)

Nationality *

12345

☆
☆
☆
☆
☆

Rank *

12345

☆
☆
☆
☆
☆

Nationality	Rank	Record	Win Streak	Knockouts, Submissions, Decisions	What features do you want in an MMA ranking	What kind of graphs would you like to display
1	4	4	4	5		bar chart
2	5	4	4	4	rank wins	
1	5	5	5	3		Pie chart
1	1	1	1	3		
5	3	5	5	5		
3	3	3	4	5		Pie chart of wins by submission, decision and ko
13	21	22	23	25		

What I have found from this method is that people don't really care about nationality and care more about how many knockouts, submissions, and decisions someone has. I think I could put losses on there, which would encompass a big record. I like the pie chart idea and the ranked wins idea.

Investigation Method 2

For my second investigation method, I will research system administrators, as they will inform me about the key requirements for the administrative side of the program, such as reliable and efficient data entry, and ease of use when inputting data into the system. I will conduct this by emailing the administrators of larger MMA statistics websites.

Questions about MMA server management

Thomas Jubb <tomjubb29@gmail.com>
to dcross

8:10 PM (7 minutes ago) ☆ 😊 ↩ ⋮

Dear DC,

My name is Tom Jubb, and I am currently working on my NEA for my A-Level Computer Science qualification, which I have chosen to undertake on the topic of MMA ranking systems. As part of this NEA, I am instructed to research what system administrators on MMA systems value. I would be extremely grateful if you could answer a few questions for me.

1. What kind of issues do admins face when they update and maintain fighter data?
2. What are the best features you like to see when using the software you use to manage fighter data?
3. From an admin perspective, what features do you commonly see missing on the software you use?

Please email me back and answer these questions.

Kind Regards,
Tom Jubb



Gregory (Tapology)

Sep 2, 2025, 22:04 GMT-5

Our software is all home-grown.

Best regards,

Greg

Tapology.com



Thomas Jubb

Sep 2, 2025, 14:13 GMT-5

...

This email is a service from Tapology. Delivered by [Zendesk](#)



Thomas Jubb

to Tapology ▾

3:08 PM (0 minutes ago)



If your software is made in-house, would you mind telling me the features you've added through the years to make it easier to use or more reliable?

...



Gregory (Tapology)

Sep 3, 2025, 09:10 GMT-5

Sorry, no.

Greg

Tapology.com

Since I have received no response from any MMA website admin, I will find Q&As online from these admins, which will have the same effect.

From what I have found, Tapology places a high value on user data validation, which can sometimes lead to frustration for organisations that submit their data to their website. However, they still have things like misspelling errors in their data. What this research has revealed to me most is a gap in user-to-admin input and communication on MMA stats websites, which underscores the need for a user voting system and a more user-oriented program.

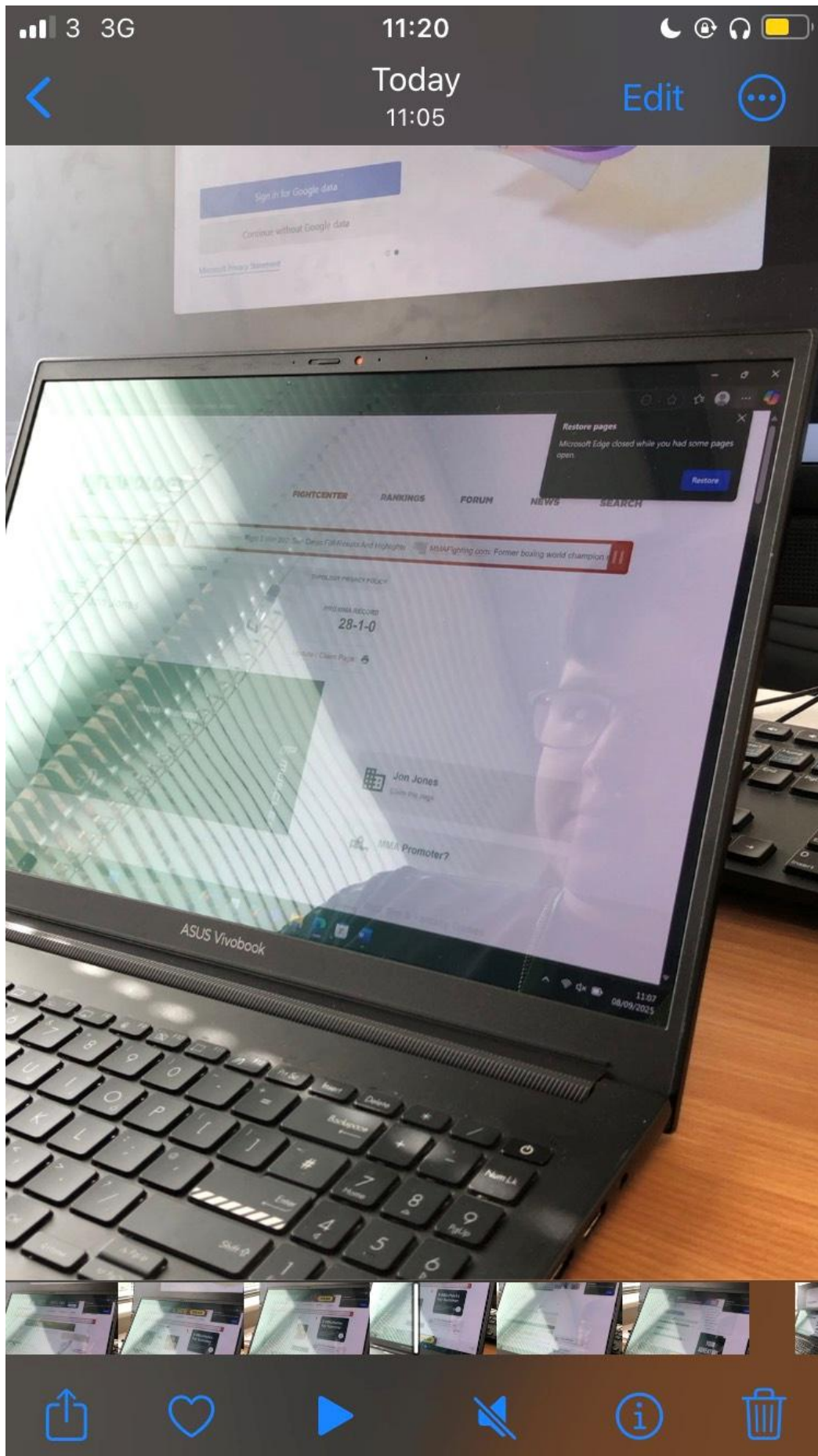
Investigation Method 3

For my third method of investigation, I want to find out more about MMA fans and how they navigate MMA websites. I can do this by observing MMA fans using an MMA website. This will tell me what people like and don't like about MMA sites in practice, what pages of the website people are most likely to look at, and what data people are most likely to find interesting.

The fan I chose first had a look at the featured fight and then looked at the individual stats of a fighter (Diego Lopez). He looked at Lopez's fighting history and upcoming fights, and from there, he looked at his upcoming opponent, who is Jean Silva. From there, he looked at the info on Jean Silva's previous fight against Bryce Mitchell. After that, he clicked the website's logo to go back to the homepage. After this, he searched for another fighter (Jon Jones), but he encountered an issue with the search because it brought him a fighter who wasn't Jon Jones.

From this observation, I have learned that MMA fans tend to go for events first and then view the context of the fight by seeing the fighters that are fighting. They also do this in reverse by looking at a fighter's fight history. They also tend to view fighter information more than fights themselves.

Because of this research, I am going to make navigation between fights as easy as possible and make sure the UI is fighter-focused. I also need to make sure that the search system is robust to stop errors like what happened with Jon Jones.



Thomas Jubb
3003

The College of Richard Collyer
Centre: 65131

Document Collection

Glicko 2

Example calculation:

Suppose a player rated 1500 competes against players rated 1400, 1550 and 1700, winning the first game and losing the next two. Assume the 1500-rated player's rating deviation is 200, and his opponents' are 30, 100 and 300, respectively. Assume the 1500 player has volatility $\sigma = 0.06$, and the system constant τ is 0.5.

Converting to the Glicko-2 scale, the player's rating and RD become 0 and 1.1513. For the opponents:

j	μ_j	ϕ_j	$g(\phi_j)$	$E(\mu, \mu_j, \phi_j)$	s_j
1	-0.5756	0.1727	0.9955	0.639	1
2	0.2878	0.5756	0.9531	0.432	0
3	1.1513	1.7269	0.7242	0.303	0

We then compute

$$\begin{aligned} v &= \left([(0.9955)^2(0.639)(1 - 0.639) \right. \\ &\quad \left. + (0.9531)^2(0.432)(1 - 0.432) + (0.7242)^2(0.303)(1 - 0.303)] \right)^{-1} \\ &= 1.7785 \end{aligned}$$

And now

$$\begin{aligned} \Delta &= 1.7785 (0.9955(1 - 0.639) + 0.9531(0 - 0.432) + 0.7242(0 - 0.303)) \\ &= -0.4834 \end{aligned}$$

The starting values in the iterative procedure are $A = a = \ln(0.06^2) = -5.62682$, and (with $k = 1$) $B = a - k\tau = -5.62682 - (1.0)(0.5) = -6.12682$, with corresponding function values computed as $f_A = -0.00053567$ and $f_B = 1.999675$. The following table summarizes the iterations (iteration 0 corresponds to the initial values above):

Iteration	A	B	f_A	f_B
0	-5.62682	-6.12682	-0.00053567	1.999675
1	-5.62682	-5.62696	-0.00026784	0.00000015238
2	-5.62696	-5.62696	0.00000015238	0.00000015238

The iterative procedure therefore converges to $A = -5.62696$, so we set $\sigma' = e^{-5.62696/2} = 0.05999$.

Now update to the new value of ϕ^* :

$$\phi^* = \sqrt{1.1513^2 + 0.05999^2} = 1.152862.$$

Next, update to the new values of ϕ' and μ' :

$$\begin{aligned} \phi' &= 1/\sqrt{\frac{1}{1.1529^2} + \frac{1}{1.7785}} = 0.8722 \\ \mu' &= 0 + (0.8722)^2 \times \\ &\quad [0.9955(1 - 0.639) \\ &\quad + 0.9531(0 - 0.432) \\ &\quad + 0.7242(0 - 0.303)] \\ &= 0 + 0.7607(-0.272) = -0.2069 \end{aligned}$$

Bradley-Terry Model

Worked example of solution procedure [\[edit \]](#)

Consider a sporting competition between four teams, who play a total of 22 games among themselves. Each team's wins are given in the rows of the table below and the opponents are given as the columns:

Results				
	A	B	C	D
A	–	2	0	1
B	3	–	5	0
C	0	3	–	1
D	4	0	3	–

For example, Team A has beat Team B twice and lost to team B three times; not played team C at all; won once and lost four times against team D.

We would like to estimate the relative strengths of the teams, which we do by calculating the parameters p_i , with higher parameters indicating greater prowess. To do this, we initialize the four entries in the parameter vector $\mathbf{p}$ arbitrarily, for example assigning the value 1 to each team: $[1, 1, 1, 1]$. Then we apply equation (5) to update p_1 , which gives

$$p_1 = \frac{\sum_{j(\neq 1)} w_{1j} p_j / (p_1 + p_j)}{\sum_{j(\neq 1)} w_{j1} / (p_1 + p_j)} = \frac{2 \frac{1}{1+1} + 0 \frac{1}{1+1} + 1 \frac{1}{1+1}}{3 \frac{1}{1+1} + 0 \frac{1}{1+1} + 4 \frac{1}{1+1}} = 0.429.$$

Now, we apply (5) again to update p_2 , making sure to use the new value of p_1 that we just calculated:

$$p_2 = \frac{\sum_{j(\neq 2)} w_{2j} p_j / (p_2 + p_j)}{\sum_{j(\neq 2)} w_{j2} / (p_2 + p_j)} = \frac{3 \frac{0.429}{1+0.429} + 5 \frac{1}{1+1} + 0 \frac{1}{1+1}}{2 \frac{1}{1+0.429} + 3 \frac{1}{1+1} + 0 \frac{1}{1+1}} = 1.172$$

Similarly for p_3 and p_4 we get

$$p_3 = \frac{\sum_{j(\neq 3)} w_{3j} p_j / (p_3 + p_j)}{\sum_{j(\neq 3)} w_{j3} / (p_3 + p_j)} = \frac{0 \frac{0.429}{1+0.429} + 3 \frac{1.172}{1+1.172} + 1 \frac{1}{1+1}}{0 \frac{1}{1+0.429} + 5 \frac{1}{1+1.172} + 3 \frac{1}{1+1}} = 0.557$$

$$p_4 = \frac{\sum_{j(\neq 4)} w_{4j} p_j / (p_4 + p_j)}{\sum_{j(\neq 4)} w_{j4} / (p_4 + p_j)} = \frac{4 \frac{0.429}{1+0.429} + 0 \frac{1.172}{1+1.172} + 3 \frac{0.557}{1+0.557}}{1 \frac{1}{1+0.429} + 0 \frac{1}{1+1.172} + 1 \frac{1}{1+0.557}} = 1.694$$

Then we normalize all the parameters by dividing by their geometric mean $(0.429 \times 1.172 \times 0.557 \times 1.694)^{1/4} = 0.830$ to get the estimated parameters $\mathbf{p} = [0.516, 1.413, 0.672, 2.041]$.

To improve the estimates further, we repeat the process, using the new $\mathbf{p}$ values. For example,

$$p_1 = \frac{2 \cdot \frac{1.413}{0.516+1.413} + 0 \cdot \frac{0.672}{0.516+0.672} + 1 \cdot \frac{2.041}{0.516+2.041}}{3 \cdot \frac{1}{0.516+1.413} + 0 \cdot \frac{1}{0.516+0.672} + 4 \cdot \frac{1}{0.516+2.041}} = 0.725.$$

Repeating this process for the remaining parameters and normalizing, we get $\mathbf{p} = [0.677, 1.034, 0.624, 2.287]$. Repeating a further 10 times gives rapid convergence toward a final solution of $\mathbf{p} = [0.640, 1.043, 0.660, 2.270]$. This indicates that Team D is the strongest and Team B the second strongest, while Teams A and C are nearly equal in strength but below Teams B and D. In this way the Bradley–Terry model lets us infer the relationship between all four teams, even though not all teams have played each other.

Mathematical details [\[edit\]](#)

Performance is not measured absolutely; it is inferred from wins, losses, and draws against other players. Players' ratings depend on the ratings of their opponents and the results scored against them. The difference in rating between two players determines an estimate for the expected score between them. Both the average and the spread of ratings can be arbitrarily chosen. The USCF initially aimed for an average club player to have a rating of 1500 and Elo suggested scaling ratings so that a difference of 200 rating points in chess would mean that the stronger player has an *expected score* of approximately 0.75.

A player's *expected score* is their probability of winning plus half their probability of drawing. Thus, an expected score of 0.75 could represent a 75% chance of winning, 25% chance of losing, and 0% chance of drawing. On the other extreme it could represent a 50% chance of winning, 0% chance of losing, and 50% chance of drawing. The probability of drawing, as opposed to having a decisive result, is not specified in the Elo system. Instead, a draw is considered half a win and half a loss. In practice, since the true strength of each player is unknown, the expected scores are calculated using the player's current ratings as follows.

If player *A* has a rating of R_A and player *B* a rating of R_B , the exact formula (using the [logistic curve](#) with base 10)^[27] for the expected score of player *A* is

$$E_A = \frac{1}{1 + 10^{(R_B - R_A)/400}}.$$

Similarly, the expected score for player *B* is

$$E_B = \frac{1}{1 + 10^{(R_A - R_B)/400}}.$$

This could also be expressed by

$$E_A = \frac{Q_A}{Q_A + Q_B}$$

and

$$E_B = \frac{Q_B}{Q_A + Q_B},$$

where $Q_A = 10^{R_A/400}$, and $Q_B = 10^{R_B/400}$. Note that in the latter case, the same denominator applies to both expressions, and it is plain that $E_A + E_B = 1$. This means that by studying only the numerators, we find out that the expected score for player *A* is Q_A/Q_B times the expected score for player *B*. We can achieve this algebraically by subtracting 1 from the reciprocal of E_B before multiplying E_B . It then follows that for each 400 rating points of advantage over the opponent, the expected score is magnified ten times in comparison to the opponent's expected score.

When a player's actual tournament scores exceed their expected scores, the Elo system takes this as evidence that player's rating is too low, and needs to be adjusted upward. Similarly, when a player's actual tournament scores fall short of their expected scores, that player's rating is adjusted downward. Elo's original suggestion, which is still widely used, was a simple linear adjustment proportional to the amount by which a player over-performed or under-performed their expected score. The maximum possible adjustment per game, called the *K-factor*, was set at $K = 16$ for masters and $K = 32$ for weaker players.

Suppose player *A* (again with rating R_A) was expected to score E_A points but actually scored S_A points. The formula for updating that player's rating is

$$R'_A = R_A + K \cdot (S_A - E_A).^{[1]}$$

This update can be performed after each game or each tournament, or after any suitable rating period. An example may help to clarify:

Suppose player *A* has a rating of 1613 and plays in a five-round tournament. They lose to a player rated 1609, draw with a player rated 1477, defeat a player rated 1388, defeat a player rated 1586, and lose to a player rated 1720. The player's actual score is $(0 + 0.5 + 1 + 1 + 0) = 2.5$. The expected score, calculated according to the formula above, was $(0.51 + 0.69 + 0.79 + 0.54 + 0.35) = 2.88$.

Therefore, the player's new rating is $[1613 + 32 \cdot (2.5 - 2.88)] = 1601$, assuming that a *K*-factor of 32 is used. Equivalently, each game the player can be said to have put an ante of *K* times their expected score for the game into a pot, the opposing player does likewise, and the winner collects the full pot of value *K*; in the event of a draw, the players [split the pot](#) and receive $\frac{1}{2}K$ points each.

Note that while two wins, two losses, and one draw may seem like a par score, it is worse than expected for player *A* because their opponents were lower rated on average. Therefore, player *A* is slightly penalized. If player *A* had scored two wins, one loss, and two draws, for a total score of three points, that would have been slightly better than expected, and the player's new rating would have been $[1613 + 32 \cdot (3 - 2.88)] = 1617$.

Processing

Odds Calculation Example – Elo System

- Made for chess, which, in my opinion, has many similarities to MMA as both are tactical, head-to-head, competitive sports. However, it is also a downside because chess can get away with having a player's experience as the main factor in a rating system, because ageing doesn't matter as much in chess as it does in MMA
- Having a predicted outcome is already integrated into the formula, which makes it simpler to make a prediction
- It is a good balance between being mathematically fair but still being rewarding to underdogs if they manage to get an upset win
- Rating changes also rely on a K-Constant to determine how big the rating change is going to be, which will mean that I will have to make a separate algorithm to determine how much a fight matters

Odds Calculation Example 2 – Glicko 2 System

- An evolution of the Glicko system, which is an evolution of the Elo system.
- Instead of relying on a K constant, Glicko 2 looks at a competitor's reliability and activity, which means that you get a more accurate model, but it still doesn't take fighter deterioration into account
- Can be computationally expensive to estimate a whole promotion's scores, but I don't think it will matter too much, as you won't have to generate the scores every time you open the program.

Odds Calculation Example 3 – Bradley-Terry Model

- Calculates the probability of a fight proportionally by the strength of each fighter, which is calculated from the fighter's past fights
- It is good for incomplete data, which is good for smaller championships
- It is easier to implement
- Only takes strength into account and no other things like Glicko 2

Data Collection Example - UFC API

- Python-based, which makes integration easier
- Easy to use, which means it will be simple to integrate this into my program
- Data may be incomplete, which is bad when we need a lot of data to calculate odds
- Little to no documentation, which means that if I have an issue, I am practically on my own

UI Example – Flask

- Is lightweight, which is better for all parties
- Works well with SQL, meaning I can integrate the database more easily.
- Works well with other Python libraries, meaning the analytics is easier to integrate
- It is easy to access for the end-user, no downloads needed if hosted on a web server; however, this also means that if I don't use a web server, Flask doesn't work out of the box.

UI Example – QT Designer

- It is very customisable, meaning that I can deploy the customisation ideas I discussed earlier more easily.
- Produce a more cross-platform desktop interface, meaning it's easier to deploy out of the box
- Has a lot of embedding support for things like graphs and SQL

Limitations

- Rankings take about 48 hours to update. This leaves fans waiting to see the impact of fights, which can be past some people's attention spans, and puts less on the table for the fighters as they won't really see the effects on the system for 48 hours.
- Some systems are cluttered with information that doesn't really matter to the casual viewer, and then also go and not show enough context, like who is a champion or not.
- Smaller promotions don't even have proper records, meaning that if you are a smaller fighter, you aren't represented.
- A lot of MMA websites don't let you filter enough data, meaning that you can't find the records you really want to find.
- A limitation that the current systems have is that they rely too much on humans, and that creates bias. The way the system in the UFC works is that 20 sports media people do a vote over the rankings. Not only does this system create bias, but it also could lead to corruption and inconsistencies in data.
- Searching can be inaccurate on some MMA websites.
- A lot of MMA websites don't value user input.

Working specification

I am going to make an MMA program with two sides. The main side will be the user side, where there will be a league table with individual fighter stats, such as nationality and record. You will be able to search for fighters based on these factors. Each fighter will have their own page, which shows their record, fight stats such as submissions and knockouts, fight history and upcoming bouts. The program will also feature an odds calculator, which will use a combination of the Glicko 2 model and the Bradley-Terry model to calculate odds. The program will also feature an entertainment predictor. The way this will work is by taking a short survey at the beginning, which will determine what the user likes or not. From this survey, the system will determine how entertaining a fight will be for you. Users will also be able to vote on fights based on entertainment.

On the admin side, you will be able to input data about fighters and fights manually through a simple interface. However, you will be able to use a UFC scraper as well.

The database will be on SQL, and the UI will use QT with a high contrast palette and simplistic visuals. The system will be heavy with visuals, with graphs such as heat maps and pie charts to show analytics about fighters.

Justifications of the methods to be used.

Firstly, for the UI, I think I will use QT Designer. I am doing this because I like how it integrates with other Python packages, and it removes an extra layer of processing that you would get with Flask, as you need to consider more server communication. This means I can allocate more resources to processing. It also means that when you get the Python programme, it works straight away, whereas with a serverless Flask approach, the user would need to worry about more things, which isn't ideal, especially if the MMA fan doesn't have much technical knowledge.

I am going to try out the UFC API with my prototype for data scraping because it seems to be the simplest way to get the job done and the best one to integrate with my SQL databases due to it being in Python, which is the primary language I am using to develop this program.

For my odds calculation. I think I am going to try to combine the Bradley-Terry model and the Glicko 2 model. That way, I can account for the lack of data, but I can add more accuracy if there is more data.

Objectives and success criteria

- The system will generate a ranked leaderboard of fighters based on the database it's given
- The leaderboard will display the fighter's name, record, nationality, weight class and methods of winning
- The table will be sortable by weight class, nationality, and the belts they've held
- You will be able to search for a fighter's name and see their page
- The fighter profile will feature past fight history, upcoming fights, statistics (wins/losses by KO, submission, decision), belts, and ranking using graphs like heat maps and pie charts
- The system must take an input of two fighters and give the odds of either of them winning by using a combination of the Glicko-2 model and the Bradley-Terry model.
- Upon first use, the system must prompt the user to make a profile so they can answer a short survey about what they find entertaining about an MMA fight. Using this, the system will be able to show how entertaining a fight is going to be to you based on what you like
- Users will also be able to vote on how entertaining a fight is going to be
- Admins will be able to manually input fighters and fights
- Admins will also have the option to use a UFC scraper.
- The system will validate inputs by rejecting duplicate data and incorrectly formatted records
- The system will use an SQL database to store data
- Rankings need to update when you enter fight results automatically
- The UI needs to be implemented with QT designer
- The user needs to be able to navigate between the league table, fighter profiles, and the odds calculator in as short a time as possible.
- The system will take a survey from the user at the beginning and make a profile that says what they find entertaining and not

- Users will be able to vote on fighters and give them a percentage approval rating
- The system will give each fighter a score according to the fighter's info to determine how entertaining the user will find the fighter.